

4-Qubit DRU-PQC Circuit Diagram

Composition with Data Re-uploading ($L = 2$) and Hybrid Classical Tail

Encoding Weights (ω)

Entangling Parameters (θ)

4D classical
input data

$$X = \begin{bmatrix} X_0 \\ X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

$|0\rangle_0$

$|0\rangle_1$

$|0\rangle_2$

$|0\rangle_3$

Data Encoding Block

$$R_X(\omega_{l,0,0}X_0 + \omega_{l,0,1})$$

$$R_X(\omega_{l,1,0}X_1 + \omega_{l,1,1})$$

$$R_X(\omega_{l,2,0}X_2 + \omega_{l,2,1})$$

$$R_X(\omega_{l,3,0}X_3 + \omega_{l,3,1})$$

Entangling Block
(Strongly Entangling Layers)

$$R_Z(\theta_{l,0,2})$$

$$R_Y(\theta_{l,0,3})$$

$$R_Z(\theta_{l,0,4})$$

$$R_Z(\theta_{l,1,2})$$

$$R_Y(\theta_{l,1,3})$$

$$R_Z(\theta_{l,1,4})$$

$$R_Z(\theta_{l,2,2})$$

$$R_Y(\theta_{l,2,3})$$

$$R_Z(\theta_{l,2,4})$$

$$R_Z(\theta_{l,3,2})$$

$$R_Y(\theta_{l,3,3})$$

$$R_Z(\theta_{l,3,4})$$

$$[\langle \sigma_z^{(0)} \rangle, \dots, \langle \sigma_z^{(3)} \rangle]$$



PyTorch Linear Layer
(4 $\rightarrow$ 2 features)

Logits

Layer l — encode-entangle motif, repeated $L = 2$ times